Color When It Counts: Grayscale-Guided Online Triggering for Always-On Streaming Video Sensing

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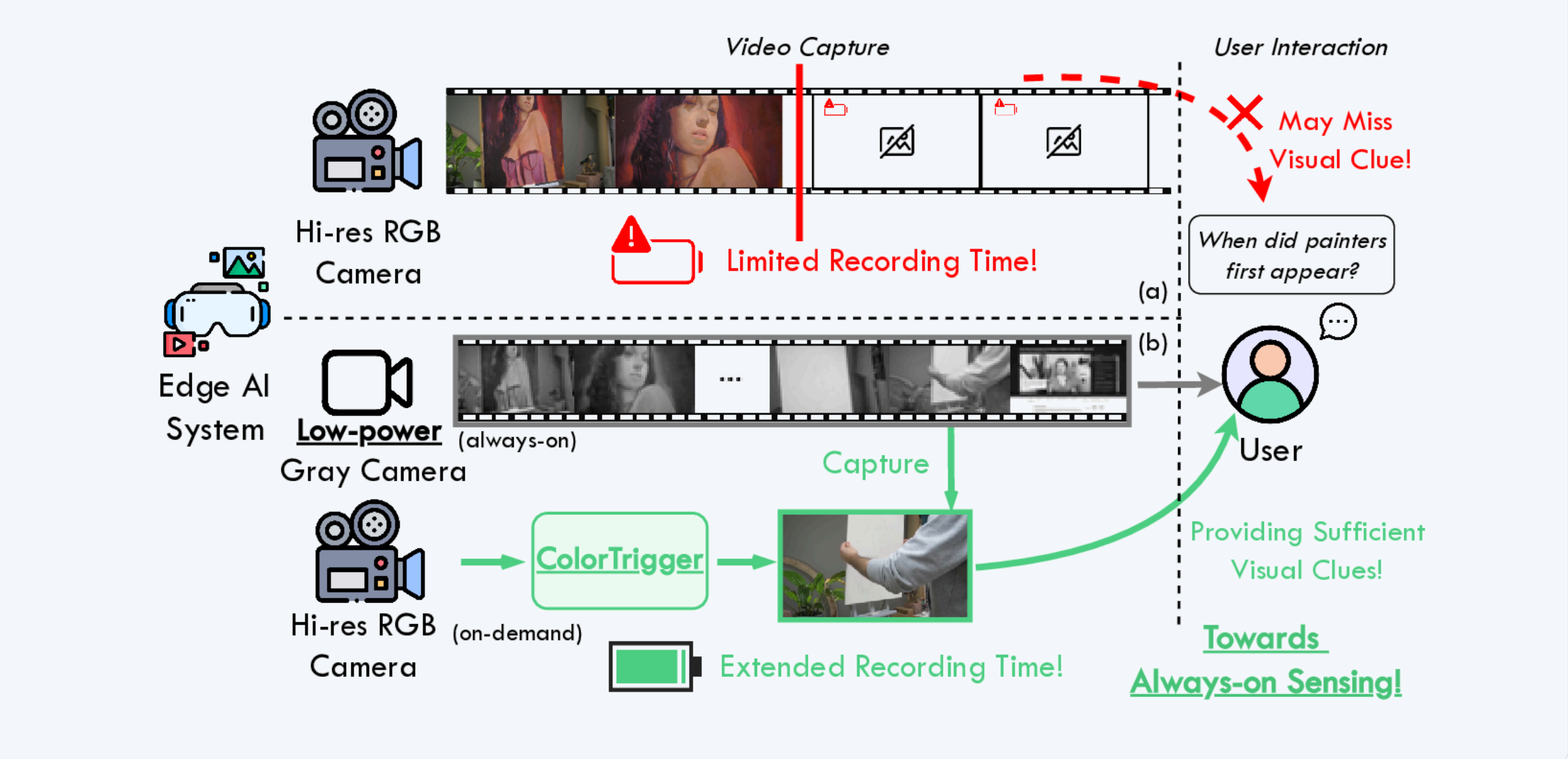
Motivation

Always-on sensing is essential for next-generation edge / wearable AI systems (smart glasses, AR, robotics).

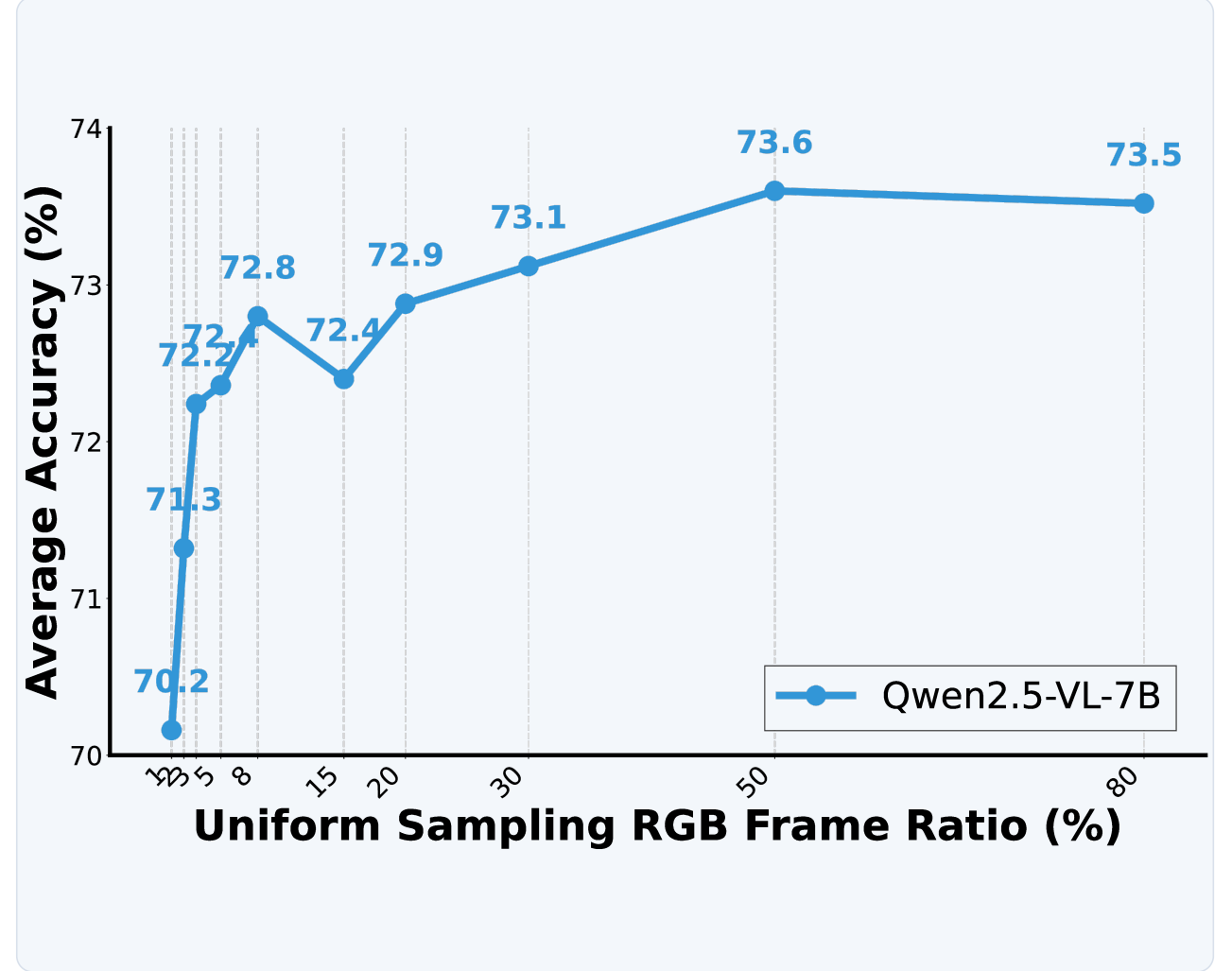
Continuous high-fidelity **RGB** capture quickly exhausts power on edge devices – the sensor → ISP → encoding → wireless pipeline lets smart glasses sustain only **30-60 min** of continuous recording, far from all-day standby.



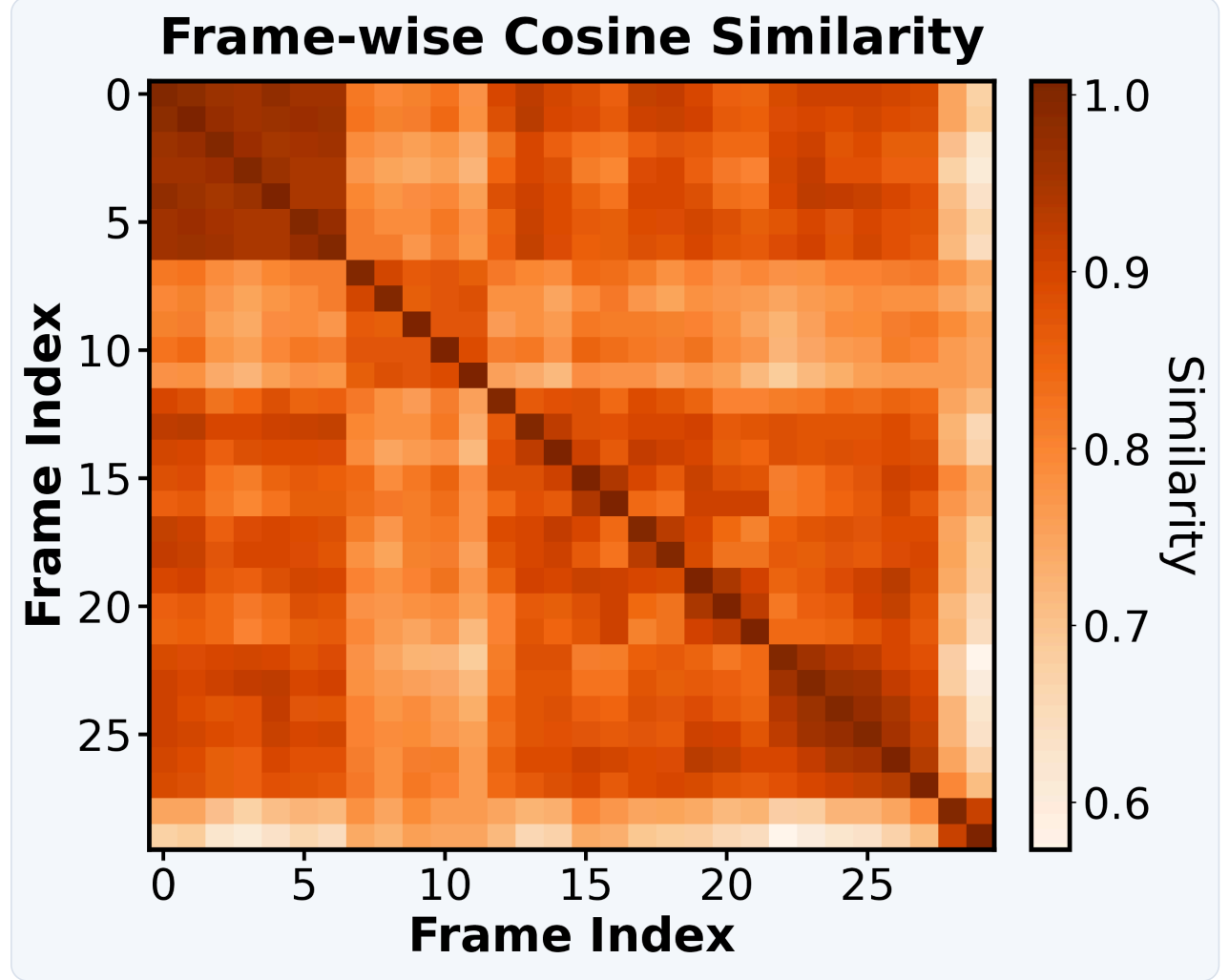
Towards always-on sensing. (a) High-res RGB pipelines drain edge power. (b) **ColorTrigger** uses a low-power grayscale camera as the always-on monitor and sparsely triggers RGB only when needed.

Key Finding

Color is not always necessary. We uniformly insert RGB frames into an otherwise grayscale stream and evaluate on StreamingBench with Qwen2.5-VL-7B: only a small fraction of RGB frames suffices. This indicates substantial color-dimension redundancy – many semantics (action recognition, layout reasoning, counting) are largely color-independent, with chromatic detail benefiting only a subset of key moments.

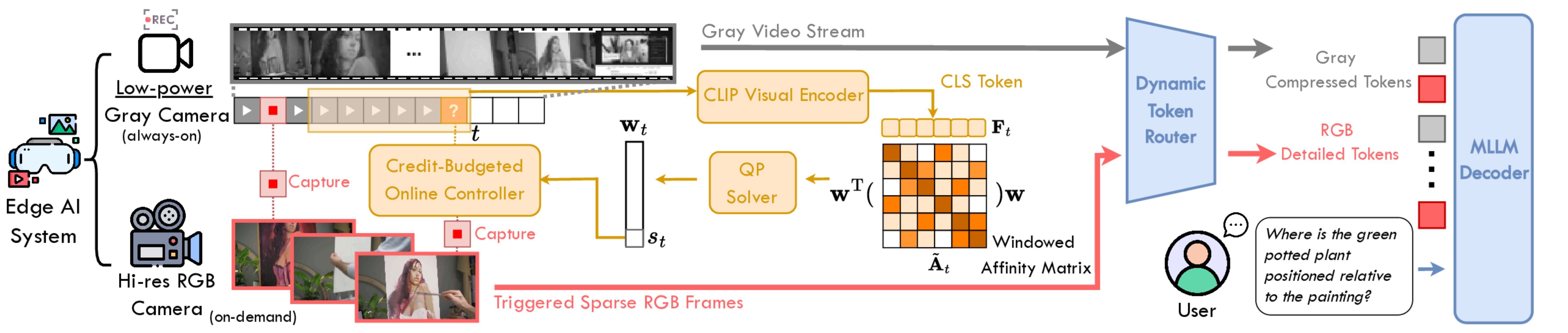


(a) Performance vs. RGB ratio. Only a small fraction of RGB frames suffices to recover comparable performance to a full-RGB baseline.



(b) Temporal attention. CLS features over 30 consecutive frames reveal significant redundancy in adjacent frames.

Method



Overview of ColorTrigger. A causal online trigger analyses a sliding window of grayscale features via an affinity matrix and a lightweight QP under a credit budget; a dynamic token router routes grayscale through a high-compression path and triggered RGB through a high-capacity path, preserving temporal order for the frozen MLLM.

ColorTrigger is a *grayscale-always, color-on-demand* framework that treats color as an on-demand signal rather than a continuously sampled modality. It comprises two components:

(i) Causal Online Trigger

- A sliding window of grayscale features is aggregated into a **windowed affinity matrix** capturing recent temporal redundancy and novelty
- A lightweight **quadratic program (QP)** decides per-frame whether color activation is warranted – strictly **causal**, no future frames
- **Credit-budgeted** controller regulates long-horizon color spending

(ii) Dynamic Token Router

- Asymmetric decoder capacity: grayscale → **64 tokens** (high-compression path); triggered RGB → **256 tokens** (high-capacity path)
- Mixed sequence reassembled in **temporal order** and fed directly to the frozen MLLM decoder
- Jointly reduces sensing and inference cost

The pipeline is **training-free**, strictly causal, and plugs into frozen MLLMs – ready for energy-constrained always-on edge deployment.

Quantitative Results

StreamingBench

Table 1. Performance comparison on StreamingBench [21] focusing on *Real-Time Visual Understanding* tasks. Real-Time Visual Understanding encompasses Object Perception (OP), Causal Reasoning (CR), Clips Summarization (CS), Attribute Perception (ATP), Event Understanding (EU), Text-Rich Understanding (TR), Prospective Reasoning (PR), Spatial Understanding (SU), Action Perception (ACP), and Counting (CT). "RGB(%)" represents the percentage of color frames remaining after triggering, where 100 indicates full color, 0 means all grayscale, and ↓ 91.9% signifies a 91.9% reduction in color frames.

Model	#Frames	RGB (%)	OP	CR	CS	ATP	EU	TR	PR	SU	ACP	CT	All
Human	-	-	89.47	92.00	93.60	91.47	95.65	92.52	88.00	88.75	89.74	91.30	91.46
Proprietary MLLMs													
Gemini 1.5 Pro [36]	1 fps	100	79.02	80.47	83.54	79.67	80.00	84.74	77.78	64.23	71.95	48.70	75.69
GPT-4o [16]	64	100	77.11	80.47	83.91	76.47	70.19	83.80	66.67	62.19	69.12	49.22	73.28
Claude 3.5 Sonnet [1]	20	100	80.49	77.34	82.02	81.73	72.33	75.39	61.11	61.79	69.32	43.09	72.44
Open-Source Video MLLMs													
LLaVA-OneVision-7B [18]	32	100	80.38	74.22	76.03	80.72	72.67	71.65	67.59	65.45	65.72	45.08	71.12
Video-LLaMA2-7B [6]	32	100	55.86	55.47	57.41	58.17	52.80	43.61	39.81	42.68	45.61	35.23	49.52
Qwen2.5-VL-7B [2]	1 fps	100	78.32	80.47	78.86	80.45	76.73	78.50	79.63	63.41	66.19	53.19	73.68
Streaming MLLMs													
Flash-VStream-7B [43]	-	100	25.89	43.57	24.91	23.87	27.33	13.08	18.52	25.20	23.87	48.70	23.23
VideoLLM-online-8B [5]	2 fps	100	39.07	40.06	34.49	31.05	45.96	32.40	31.48	34.16	42.49	27.89	35.99
Dispider-7B [27]	1 fps	100	74.92	75.53	74.10	73.08	74.44	59.92	76.14	62.91	62.16	45.80	67.63
TimeChat-Online-7B [41]	1 fps	100	80.22	82.03	79.50	83.33	76.10	78.50	78.70	64.63	69.60	57.98	75.36
InternVL-3.5-8B	128	100	83.47	82.03	82.65	84.62	75.47	80.06	81.48	67.89	70.45	59.04	77.20
InternVL-3.5-8B grayscale	128	0	60.98	78.91	60.88	55.45	65.41	63.24	75.00	60.16	62.22	55.85	62.08
ColorTrigger (Ours, 8B)	128	8.1 (↓91.9%)	75.07	77.34	72.56	75.64	71.70	70.40	76.85	65.04	66.48	57.98	70.72 (↑8.64%)
ColorTrigger (Ours, 8B)	128	34.3 (↓65.7%)	81.57	78.91	77.29	84.94	76.10	77.57	81.48	67.48	67.61	56.91	75.24 (↑13.16%)

- Achieves **91.6%** of the full-color baseline performance while using only **8.1%** RGB frames
- Matches Gemini 1.5 Pro (75.69) and beats GPT-4o (73.28); **+8.64** over the grayscale-only baseline

OVO-Bench

Table 2. Results on OVO-Bench [25] comprising three categories: i) *Real-Time Visual Perception* (OCR: Optical Character Recognition, ACR: Action Recognition, ATR: Attribute Recognition, STU: Spatial Understanding, FPD: Future Prediction, OJR: Object Recognition), ii) *Backward Tracing* (EPM: Episodic Memory, ASI: Action Sequence Identification, HLD: Hallucination Detection), and iii) *Forward Active Responding* (REC: Repetition Event Count, SSR: Sequential Steps Recognition, CRR: Clues Reveal Responding).

Model	#Frames	RGB (%)	Real-Time Visual Perception							Backward Tracing				Forward Active Responding				Overall
			OCR	ACR	ATR	STU	FPD	OJR	Avg.	EPM	ASI	HLD	Avg.	REC	SSR	CRR	Avg.	Avg.
Human Agents	-	-	94.0	92.6	94.8	92.7	91.1	94.0	93.2	92.6	93.0	91.4	92.3	95.5	89.7	93.6	92.9	92.8
Proprietary MLLMs																		
Gemini 1.5 Pro [36]	1 fps	100	87.3	67.0	80.2	54.5	68.3	67.4	70.8	68.6	75.7	52.7	62.3	35.5	74.2	61.7	57.2	65.3
GPT-4o [16]	64	100	69.1	65.1	65.5	50.0	68.3	63.7	63.6	49.8	71.0	55.4	58.7	27.6	73.2	59.4	53.4	58.6
Open-Source Video MLLMs																		
LLaVA-NeXT-Video-7B [47]	64	100	69.8	59.6	66.4	50.6	72.3	61.4	63.3	51.2	64.2	9.7	41.7	34.1	67.6	60.8	54.2	53.1
LLaVA-OneVision-7B [18]	64	100	67.1	58.7	69.8	49.4	71.3	60.3	62.8	52.5	58.8	23.7	45.0	24.8	66.9	60.8	50.9	52.9
Qwen2-VL-7B [37]	64	100	69.1	53.2	63.8	50.6	66.3	60.9	60.7	44.4	66.9	34.4	48.6	30.1	65.7	50.8	48.9	52.7
LongVU-7B [32]	1 fps	100	55.7	49.5	59.5	48.3	68.3	63.0	57.4	43.1	66.2	9.1	39.5	16.6	69.0	60.0	48.5	48.5
Streaming MLLMs																		
Flash-VStream-7B [43]	1 fps	100	25.5	32.1	29.3	33.7	29.7	28.8	29.9	36.4	33.8	5.9	25.4	5.4	67.3	60.0	44.2	33.2
VideoLLM-online-8B [5]	2 fps	100	8.1	23.9	12.1	14.0	45.5	21.2	20.8	22.2	18.8	12.2	17.7	-	-	-	-	-
TimeChat-Online-7B [41]	1 fps	100	75.2	46.8	70.7	47.8	69.3	61.4	61.9	55.9	59.5	9.7	41.7	31.6	38.5	40.0	36.7	46.7
Qwen2.5-VL-7B [2]	1 fps	100	73.8	56.0	68.1	46.6	71.3	60.3	62.7	48.2	64.9	26.9	46.6	36.2	41.8	47.1	41.7	50.3
InternVL-3.5-8B	128	100	77.9	64.2	75.9	56.7	75.3	70.7	70.1	56.6	66.2	34.4	52.4	48.2	62.7	41.3	50.7	57.7
InternVL-3.5-8B grayscale	128	0	61.1	46.8	53.5	47.8	56.4	57.1	53.8	46.1	54.7	37.1	46.0	35.3	54.3	42.1	43.9	47.9
ColorTrigger (Ours, 8B)	128	7.1 (↓92.9%)	65.1	52.3	61.2	50.6	66.3	63.6	59.9 (↑6.1%)	50.5	58.1	32.3	47.0	32.5	59.3	41.3	44.3	50.4 (↑2.5%)
ColorTrigger (Ours, 8B)	128	33.1 (↓66.9%)	77.9	56.0	69.8	55.1	67.3	65.2	65.2 (↑11.4%)	50.2	62.8	30.7	41.3	32.7	59.4	40.8	44.3	52.5 (↑4.6%)

- Achieves overall **52.5** with only 33.1% RGB – beats almost all open-source online MLLMs
- Real-Time Visual Perception jumps **+11.4** points over the grayscale-only baseline

Video-MME

Table 3. Performance comparison on Video-MME [10].

Method	#Frames	RGB (%)	Video-MME			
			short	medium	long	overall
VideoChat2-7B [19]	16	100	48.3	37.0	33.2	39.5
LongVA-7B [46]	128	100	61.1	50.4	46.2	52.6
Kangaroo-7B [22]	64	100	66.1	55.3	46.6	56.0
Video-CCAM-14B [9]	96	100	62.2	50.6	46.7	53.2
VideoXL-7B [33]	128	100	64.0	53.2	49.2	55.5
Dispider-7B [27]	1 fps	100	-	-	-	57.2
VideoChat-Online-4B [15]	2 fps	100	-	-	47.1	54.4
TimeChat-Online-7B [41]	1 fps	100	-	-	48.4	62.4
InternVL-3.5-8B	128	100	76.7	65.3	54.7	65.6
InternVL-3.5-8B grayscale	128	0	63.4	57.9	50.6	57.3
ColorTrigger (Ours, 8B)	128	9.1 (↓90.9%)	71.9	63.0	53.6	62.8 (↑5.5%)
ColorTrigger (Ours, 8B)	128	37.6 (↓62.4%)	76.8	66.3	55.1	66.1 (↑8.8%)

- Surpasses the full-RGB InternVL-3.5-8B baseline (65.6) with overall **66.1** using 37.6% RGB
- Uses **62.4% fewer** chromatic frames than the full-RGB baseline

Ablation

Component study. Full model recovers **91.6%** at 8.1% RGB. −Grayscale → −1.96; uniform sampling → −1.72.

Table 4. Component Study. Δ indicates the relative performance (%) w.r.t. baseline. UniRGB and TrigRGB represent uniformly sampled and triggered RGB frames, respectively.

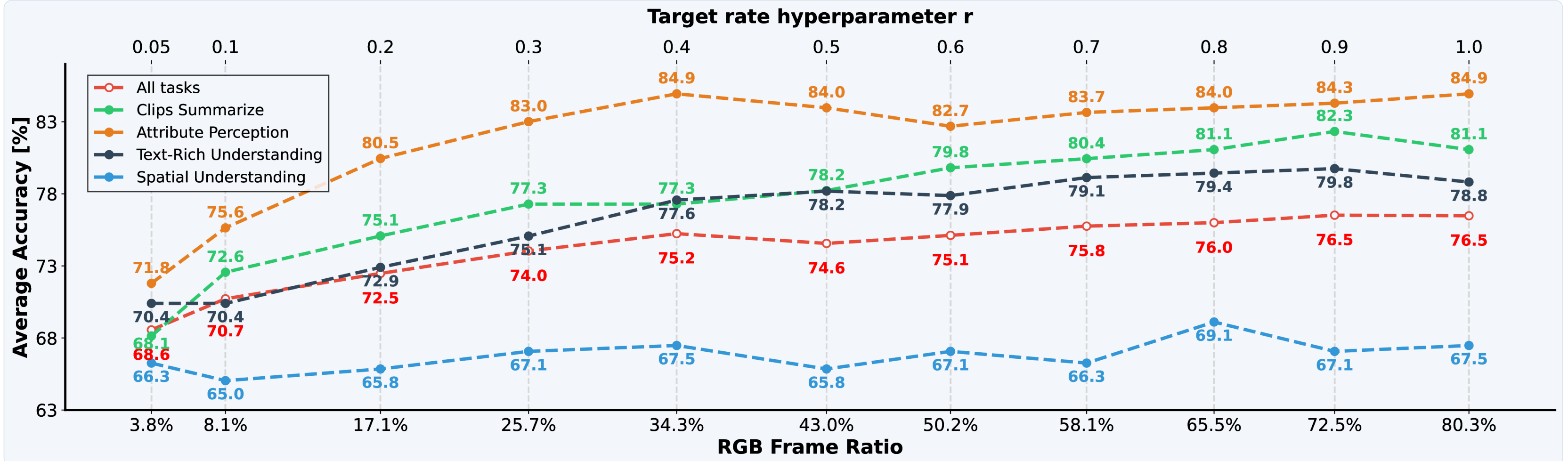
Model	RGB (%)	All	Δ
Baseline	100	77.20	100.0%
Grayscale	0	62.08	80.4%
Only UniRGB	8.1	65.52	84.9%
Grayscale + UniRGB	8.1	69.00	89.4%
Only TrigRGB	8.1	68.76	89.1%
Grayscale + TrigRGB (Ours)	8.1	**70.72**	**91.6%**
Only UniRGB	34.3	70.84	91.8%
Grayscale + UniRGB	34.3	72.64	94.1%
Only TrigRGB	34.3	73.48	95.2%
Grayscale + TrigRGB (Ours)	34.3	**75.24**	**97.5%**

Token efficiency. Dynamic Token Router → **31.1%** tokens at 91.6% recovery; 50.7% tokens at 75.24.

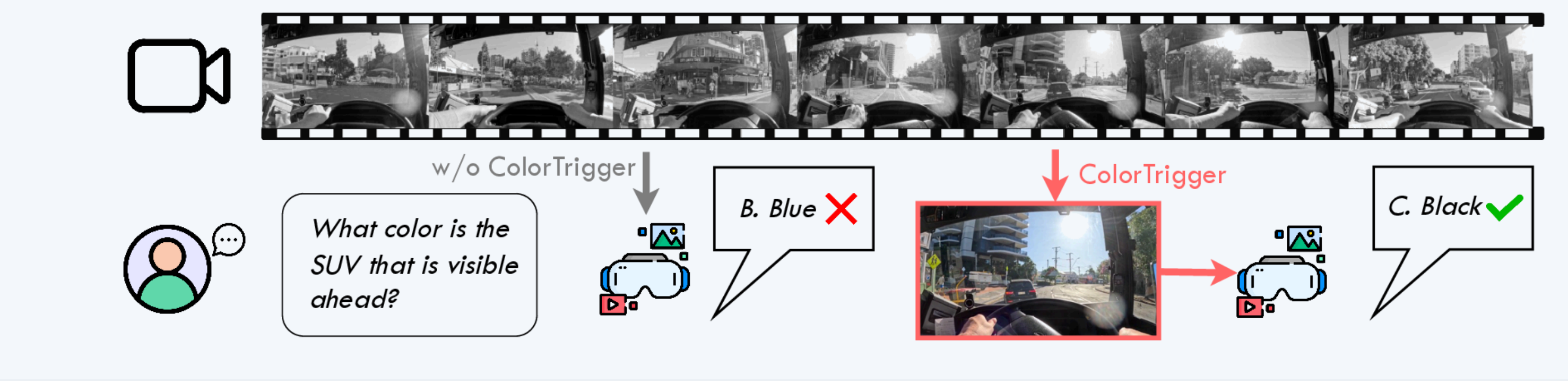
Table 5. Token Efficiency Analysis. Δ denotes the percentage relative to baseline. DT represents Dynamic Token Router, and TrigRGB represents triggered RGB frame selection.

Model	Gray Tokens/Frame	RGB(%)	All	#VisualToken	Δ
Full RGB	-	100	77.20	100.0%	100.0%
Gray	256	0	68.76	100.0%	89.1%
Gray	64	0	62.08	25.0%	80.4%
Gray + TrigRGB	256	8.1	73.68	100.0%	95.4%
Gray + TrigRGB	64	8.1	67.08	25.0%	86.9%
Gray + TrigRGB + DT	64	8.1	70.72	**31.1%**	91.6%
Gray + TrigRGB	256	34.3	76.32	100.0%	98.9%
Gray + TrigRGB	64	34.3	68.96	25.0%	89.3%
Gray + TrigRGB + DT	64	34.3	75.24	**50.7%**	97.5%

Performance across varying RGB frame ratios. As *r* grows from 0.05 to 1.0, accuracy rises monotonically – *color is beneficial but not always necessary*. Task dependencies vary: Attribute Perception saturates at ~26% RGB; Spatial Understanding stays flat (65–69%); Clips Summarization and Text-Rich Understanding gain gradually.



Qualitative Example



Grayscale-only mispredicts SUV color as *Blue*; **ColorTrigger** triggers RGB and answers **Black**.

Conclusion

- **Color is beneficial but not always necessary** – sparse RGB on a continuous grayscale stream is enough.
- **Causal, training-free, edge-ready**: practical always-on video sensing on resource-constrained devices.
- Selective triggering at critical moments can **match or even exceed** full-RGB performance.

Paper: arxiv.org/abs/2603.22466 · **Code:** github.com/lvgd/ColorTrigger · **Project:** weitongcai.com/ColorTrigger